

Hynes Birmingham

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EDUCATION

University of Connecticut

B.S. Biomedical Engineering | Minor in Computer Science | Concentration: Systems & Instrumentation

GPA: 3.41 | CAPS Research Scholar

Storrs, CT

May 2029

RESEARCH & ENGINEERING EXPERIENCE

Ryberg Lab — University of Connecticut, Allied Health Sciences

Undergraduate Researcher — CAPS Research Scholar

Storrs, CT

Nov 2025 – Present

- Designed and built a continuous, self-validating testing pipeline for a real-time biomedical imaging system: synthetic data generation, automated test execution, provenance-tracked validation reporting; 59/59 unit tests passing, 6,345 SLOC
- Built a two-layer anomaly detection system combining a statistical detector (Z-score/IQR) with a Random Forest classifier; pipeline achieves 96.3% count accuracy and F1 0.957 on 820-plate synthetic validation set
- Structured 39-organism $\times$ 11-media-type profile system with per-run provenance tracking (git commit hash, environment snapshot, input hash); accepted to present original research at UMBC Undergraduate Research Conference (Sep 2026)

FocusGuard — Multi-Modal Sensor Fusion System

Jan 2026 – Present

- Architected a multi-threaded producer-consumer system fusing real-time computer vision (MediaPipe face mesh) with keyboard behavioral signals; 26 ms median sensor-to-score latency at 30 fps, 116 MB steady-state RSS, zero crashes across 6-minute soak (10/10 unit tests, 1,008 SLOC)
- Thread-safe lock-guarded shared state across independently-clocked sensor threads; Flask REST API and live dashboard for real-time and historical data

TECHNICAL PROJECTS

Automated Bacterial Colony Counter — Full Stack Instrument

Feb 2026 – Present

- Python/OpenCV CV pipeline (96.3% count accuracy, F1 0.957, <0.10% false-positive rate) deployed as a Flask web application on embedded Linux (Raspberry Pi 5); end-to-end API response 71.1 ± 24.6 ms, 0 failures across 120 timed runs
- Custom power-control PCB designed in KiCad and ordered; 62.3 ± 0.8 ms per-plate processing time

HuskyBook — Peer-to-Peer Campus Directory

Jan 2026 – Present

- Full-stack web app: Node.js + Express REST API with 16 endpoints across 5-table SQLite schema, Vite front end, magic-link auth; 0.07–0.16 ms read latency, 10,600 req/s throughput, 32/32 tests passing, 1,724 SLOC

STM32 Bare-Metal GPIO Driver

Jan 2026 – Present

- Register-level GPIO driver for STM32F411RE in C; 9 MMIO registers across 3 peripherals, no HAL abstraction, 752 B flash image, logic analyzer verification

Pharmacokinetic Simulator

Nov 2025 – Dec 2025

- Two-compartment ODE model in Python; RK45 solver validated against closed-form analytic solution to <0.001% error on AUC and $t_{1/2}$; 3.03 ms single-dose runtime

LEADERSHIP

BRIDGE STEM Coach — Vergnano Institute of Impact, UConn

Jun 2026 – Aug 2026

- Coached 43 incoming first-generation and underrepresented STEM students through UConn's BRIDGE transition program
- Independently designed and delivered a biotechnology and instrumentation module to a cohort of 8 students covering bacterial transformation, fluorescence imaging, and quantitative data analysis
- Built a companion Python tool in Google Colab enabling students without local software access to fully process and visualize their own experimental data

Learning Community Tutor — Scholars House, UConn

Fall 2025 – Spring 2026

- Provided STEM tutoring to residents of Scholars House, a living-learning community for Black men in higher education; 2–3 sessions/week over 12 weeks, contributing to the community's 3.0 GPA goal by Spring 2026

Innovation Zone — University of Connecticut

Students as Makers (SAMs) — Founding Cohort

Storrs, CT

Feb 2025 – Dec 2025

- Member of the founding cohort of UConn's Students as Makers program; fabricated 7 projects for improvement of the Innovation Zone and related services utilizing 3D printing, manufacturing, and power tools, saving over \$2,300 compared to purchasing equivalent equipment or products

Jersey Mike's Subs

Shift Manager

West Hartford, CT

Nov 2025 – Present

- Process over \$4,000 in weekly cash sales; coordinate teams of 5–8 employees in a high-volume environment

SKILLS

Programming: Python, C, JavaScript, MATLAB

Software Engineering: OOP design, Flask/REST APIs, Node.js + Express, multi-threaded systems, automated testing, Git, SQLite

Data & ML: ETL pipelines, scikit-learn, pandas, NumPy, OpenCV, MediaPipe, provenance tracking

Hardware & Embedded: STM32/ARM Cortex-M4, register-level C drivers, KiCad PCB design, ARM GCC, embedded Linux, logic analyzer